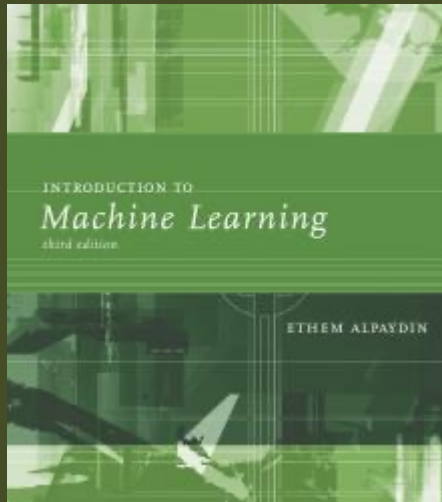




Lecture Slides for

INTRODUCTION TO MACHINE LEARNING

3RD EDITION

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CHAPTER 4:

PARAMETRIC METHODS

Parametric Estimation

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□ $\mathcal{X} = \{x^t\}_t$ where $x^t \sim p(x)$

□ Parametric estimation:

Assume a form for $p(x \mid \theta)$ and estimate θ , its sufficient statistics, using X

e.g., $N(\mu, \sigma^2)$ where $\theta = \{\mu, \sigma^2\}$

Maximum Likelihood Estimation

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- Likelihood of θ given the sample $\mathcal{X}$

$$l(\vartheta | \mathcal{X}) = p(\mathcal{X} | \vartheta) = \prod_t p(x^t | \vartheta)$$

- Log likelihood

$$\mathcal{L}(\vartheta | \mathcal{X}) = \log l(\vartheta | \mathcal{X}) = \sum_t \log p(x^t | \vartheta)$$

- Maximum likelihood estimator (MLE)

$$\vartheta^* = \operatorname{argmax}_{\vartheta} \mathcal{L}(\vartheta | \mathcal{X})$$

Examples: Bernoulli/Multinomial

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- **Bernoulli:** Two states, failure/success, x in $\{0,1\}$

$$P(x) = p_o^x (1 - p_o)^{(1-x)}$$

$$\mathcal{L}(p_o | \mathcal{X}) = \log \prod_t p_o^{x^t} (1 - p_o)^{(1-x^t)}$$

$$\text{MLE: } p_o = \sum_t x^t / N$$

- **Multinomial:** $K > 2$ states, x_i in $\{0,1\}$

$$P(x_1, x_2, \dots, x_K) = \prod_i p_i^{x_i}$$

$$\mathcal{L}(p_1, p_2, \dots, p_K | \mathcal{X}) = \log \prod_t \prod_i p_i^{x_i^t}$$

$$\text{MLE: } p_i = \sum_t x_i^t / N$$

Bernoulli

$$x \in \{0, 1\}$$

$$P(X=1) = p \quad P(X=0) = 1-p$$

$$P(X=x) = p^x (1-p)^{1-x} \quad \text{for } x \in \{0, 1\}.$$

Data: $X = \{x^t\}_{t=1}^N$ $x^t \in \{0, 1\}$. e.g., tails, heads $X = \{T, T, H, H, T, H, H, \dots\}$
 $p \equiv \text{prob of Heads.}$

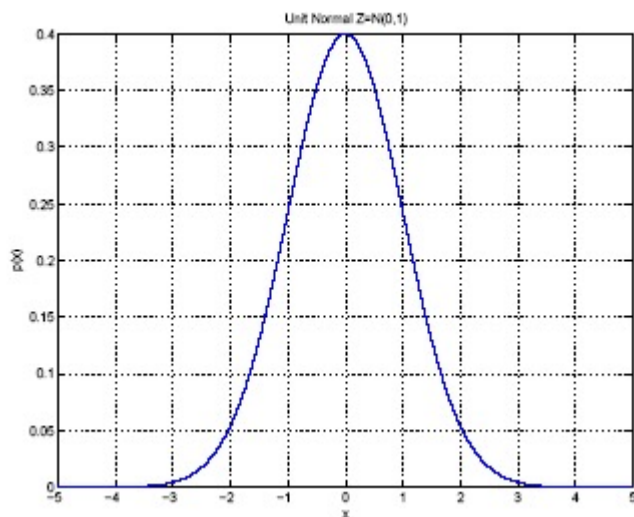
$$\ell(p|X) = \prod_t p^{x^t} (1-p)^{1-x^t}$$

$$\ell(p|X) = \log \prod_t p^{x^t} (1-p)^{1-x^t} = \sum_t x^t \log p + (1-x^t) \log (1-p)$$

$$\frac{d\ell}{dp} = 0 \Rightarrow \sum_{t=1}^N \frac{x^t}{p} - \frac{(1-x^t)}{1-p} = 0 \Rightarrow \sum_t x^t - \cancel{p^{x^t}} - p + \cancel{p^{x^t}} = 0$$
$$\sum_t x^t = N \cdot p \Rightarrow p = \frac{\sum_t x^t}{N}$$

MLE

Gaussian (Normal) Distribution



μ σ

$$m = \frac{\sum_t x^t}{N}$$

$$s^2 = \frac{\sum_t (x^t - m)^2}{N}$$

□ $p(x) = \mathcal{N}(\mu, \sigma^2)$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

□ MLE for μ and σ^2 :

Gaussian Density

$$p(x) \sim N(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$



$$X = \{x^t\}_{t=1}^N \quad x^t \sim N(\mu, \sigma^2)$$

$$\ell(\mu, \sigma^2 | X) = \prod_t p(x^t | \mu, \sigma^2) = \prod_t \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x^t - \mu)^2}{2\sigma^2}\right]$$

$$\mathcal{L}(\mu, \sigma^2 | X) = \log \ell(\mu, \sigma^2 | X) = \sum_t \left[-\frac{1}{2} \log 2\pi - \log \sigma - \frac{(x^t - \mu)^2}{2\sigma^2} \right]$$

$$= -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{1}{2\sigma^2} \sum_t (x^t - \mu)^2$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = 0 \Rightarrow \sum_t (x^t - \mu) = 0 \Rightarrow \mu = \frac{\sum_t x^t}{N} \quad \frac{\partial \mathcal{L}}{\partial \sigma} = 0 \Rightarrow -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_t (x^t - \mu)^2 = 0$$
$$\sigma^2 = \frac{\sum_t (x^t - \mu)^2}{N}$$

Bias and Variance

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Unknown parameter θ

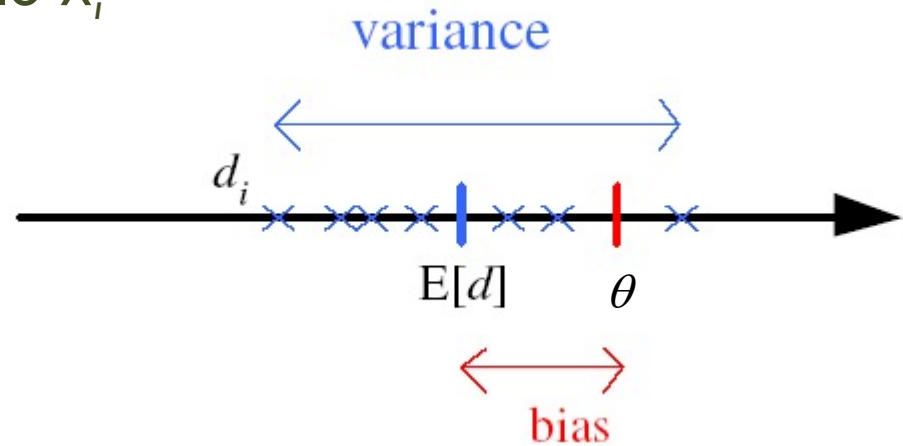
Estimator $d_i = d(X_i)$ on sample X_i

Bias: $b_\theta(d) = E[d] - \theta$

Variance: $E[(d - E[d])^2]$

Mean square error:

$$\begin{aligned} r(d, \theta) &= E[(d - \theta)^2] \\ &= (E[d] - \theta)^2 + E[(d - E[d])^2] \\ &= \text{Bias}^2 + \text{Variance} \end{aligned}$$



Parametric Classification

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$$g_i(x) = p(x | C_i) P(C_i)$$

or

$$g_i(x) = \log p(x | C_i) + \log P(C_i)$$

$$p(x | C_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right]$$

$$g_i(x) = -\frac{1}{2} \log 2\pi - \log \sigma_i - \frac{(x - \mu_i)^2}{2\sigma_i^2} + \log P(C_i)$$

□ Given the sample $\mathcal{X} = \{x^t, r^t\}_{t=1}^N$

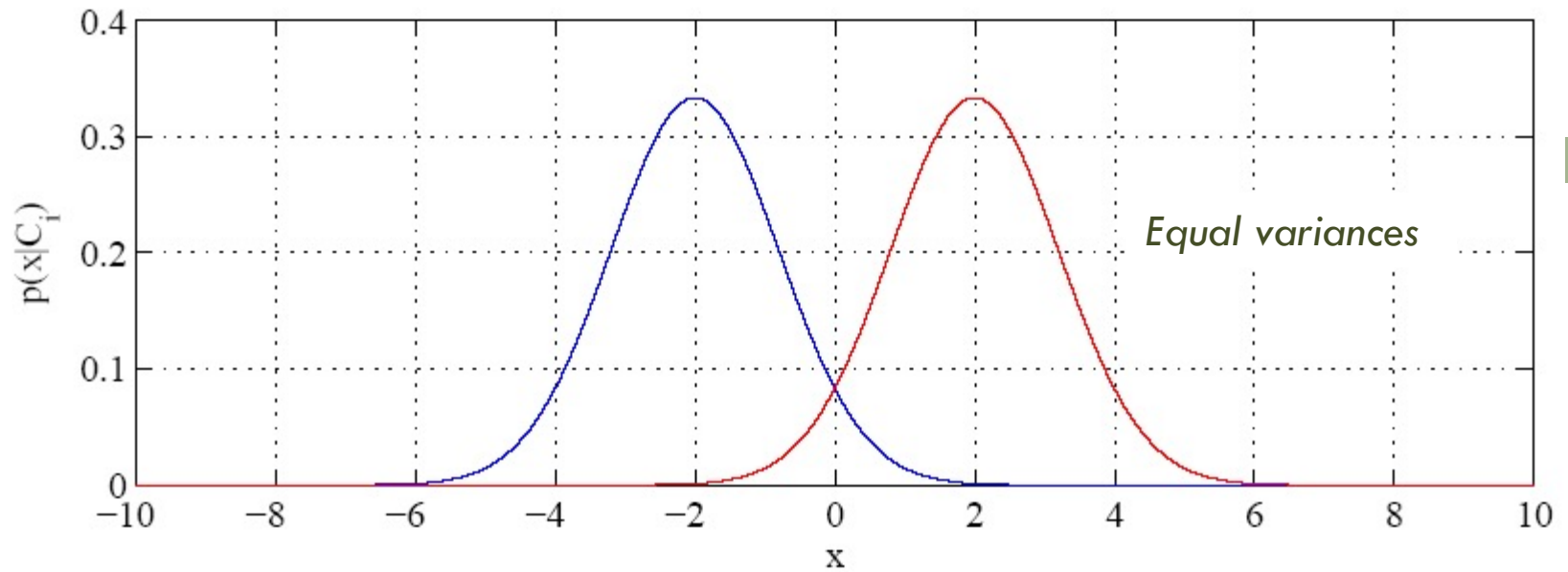
$$x \in \mathfrak{R} \quad r_i^t = \begin{cases} 1 & \text{if } x^t \in C_i \\ 0 & \text{if } x^t \in C_j, j \neq i \end{cases}$$

□ ML estimates are

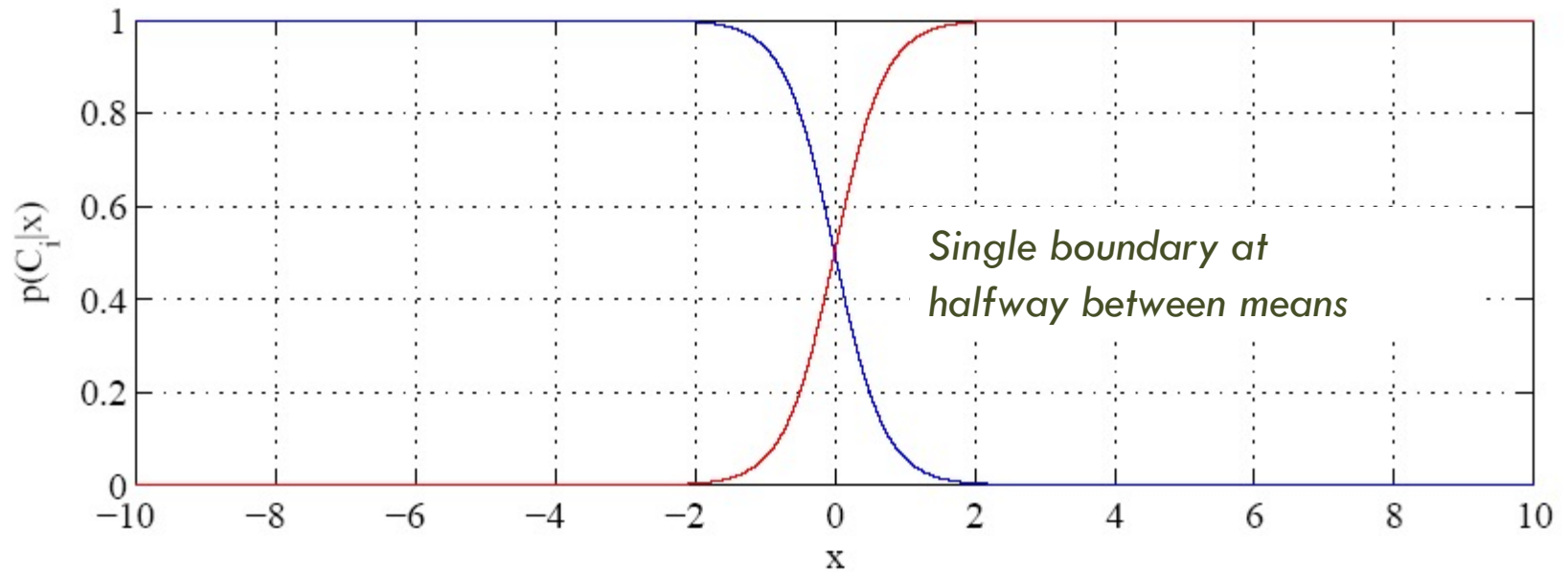
$$\hat{P}(C_i) = \frac{\sum_t r_i^t}{N} \quad m_i = \frac{\sum_t x^t r_i^t}{\sum_t r_i^t} \quad s_i^2 = \frac{\sum_t (x^t - m_i)^2 r_i^t}{\sum_t r_i^t}$$

□ Discriminant $g_i(x) = -\frac{1}{2} \log 2\pi - \log s_i - \frac{(x - m_i)^2}{2s_i^2} + \log \hat{P}(C_i)$

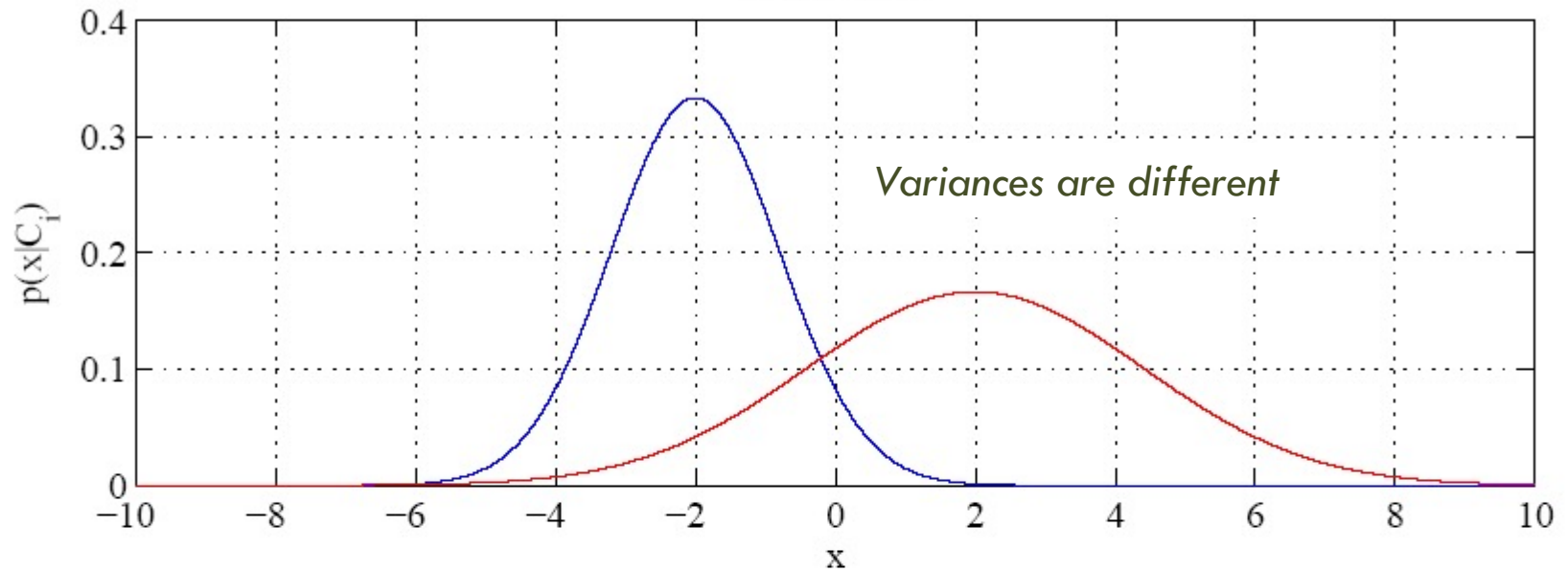
Likelihoods



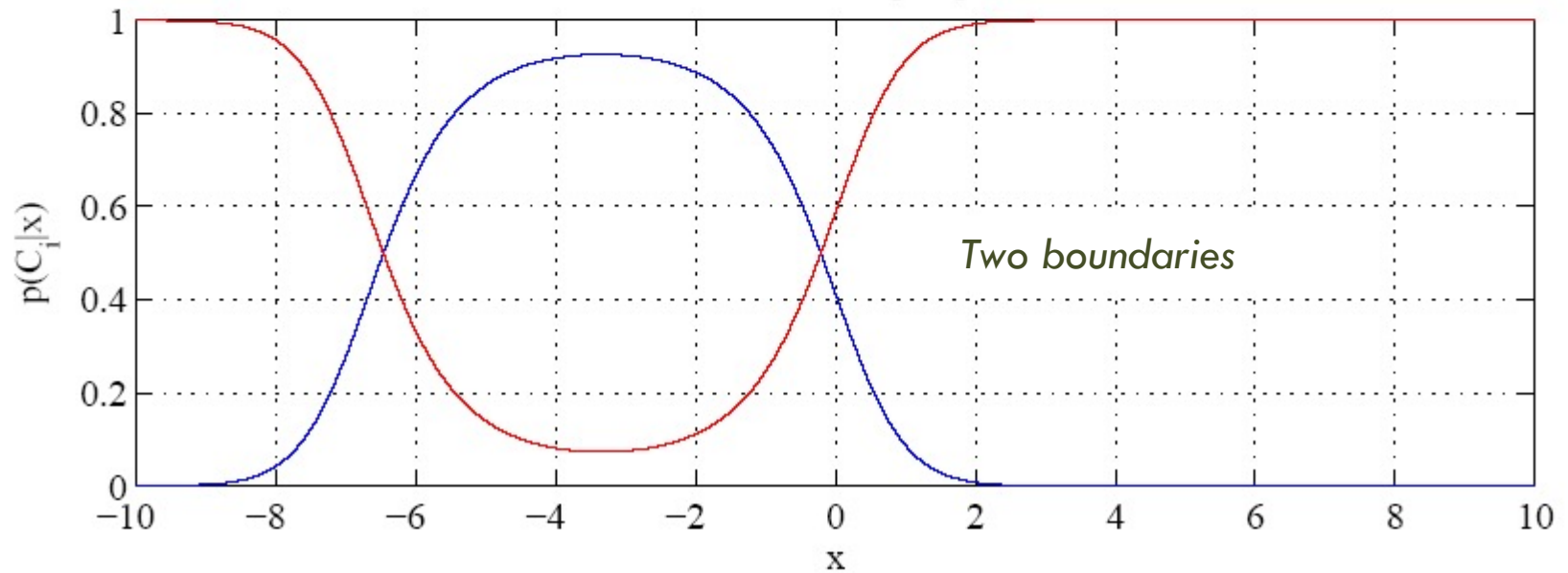
Posteriors with equal priors

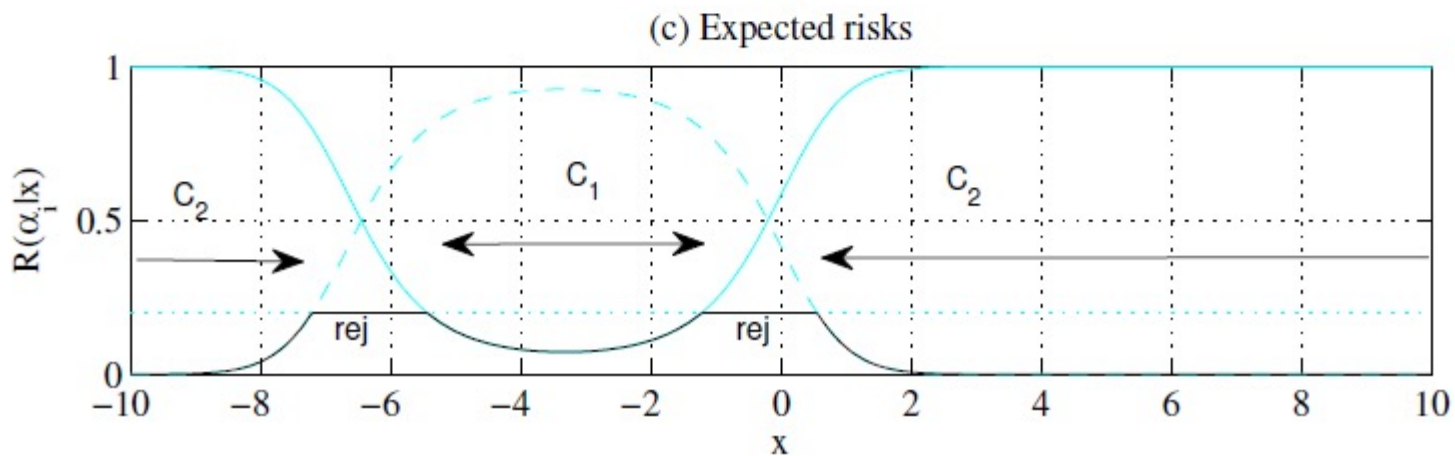
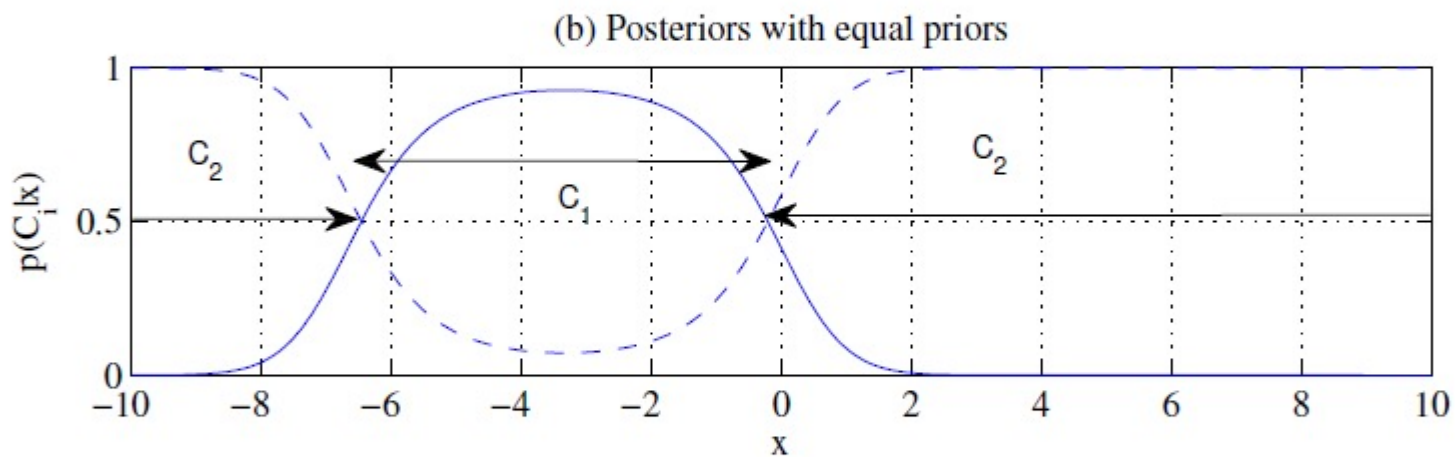
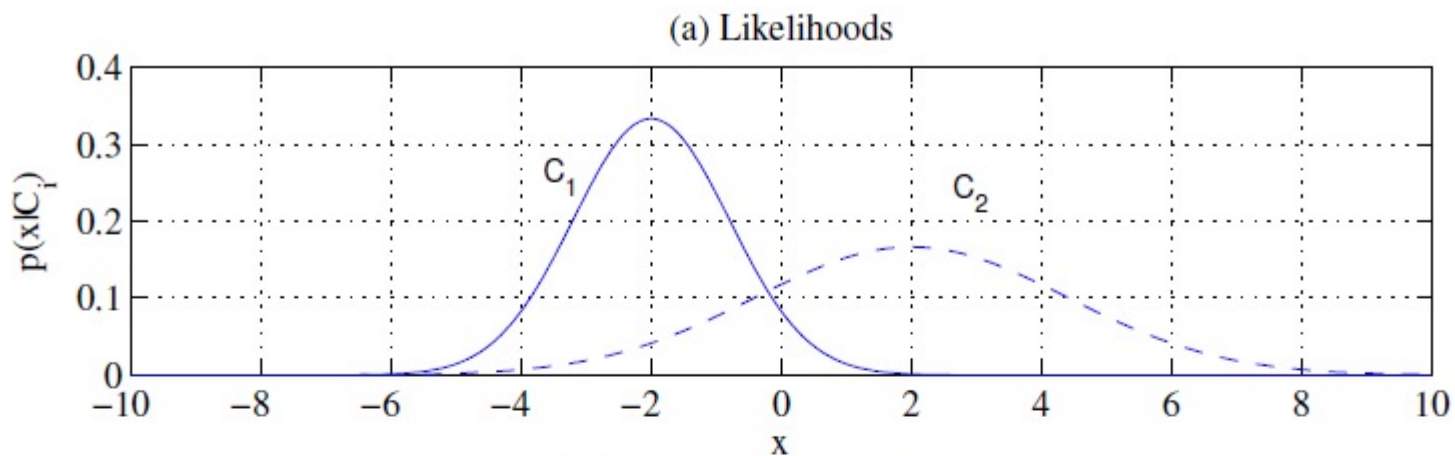


Likelihoods



Posteriors with equal priors





Regression

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$$r = f(x) + \varepsilon$$

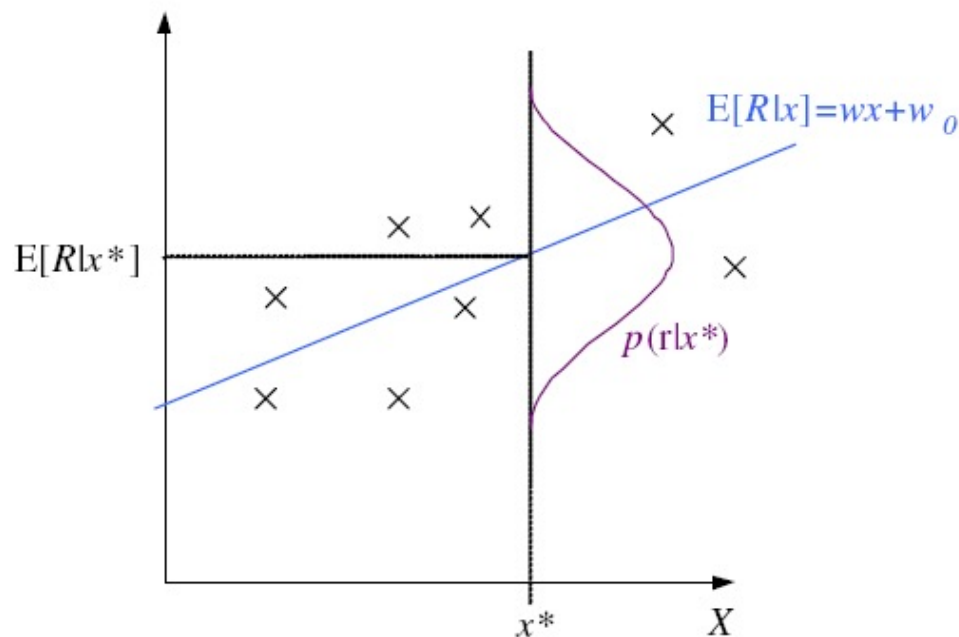
$$\text{estimator: } g(x | \theta)$$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$p(r | x) \sim \mathcal{N}(g(x | \theta), \sigma^2)$$

$$\mathcal{L}(\theta | \mathcal{X}) = \log \prod_{t=1}^N p(x^t, r^t)$$

$$= \log \prod_{t=1}^N p(r^t | x^t) + \log \prod_{t=1}^N p(x^t)$$



Regression: From LogL to Error

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$$\begin{aligned}\mathcal{L}(\theta | \mathcal{X}) &= \log \prod_{t=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{[r^t - g(x^t | \theta)]^2}{2\sigma^2} \right] \\ &= -N \log \sqrt{2\pi}\sigma - \frac{1}{2\sigma^2} \sum_{t=1}^N [r^t - g(x^t | \theta)]^2 \\ E(\theta | \mathcal{X}) &= \frac{1}{2} \sum_{t=1}^N [r^t - g(x^t | \theta)]^2\end{aligned}$$

Linear Regression

$$g(x^t | w_1, w_0) = w_1 x^t + w_0$$

$$\sum_t r^t = N w_0 + w_1 \sum_t x^t$$

$$\sum_t r^t x^t = w_0 \sum_t x^t + w_1 \sum_t (x^t)^2$$

$$\mathbf{A} = \begin{bmatrix} N & \sum_t x^t \\ \sum_t x^t & \sum_t (x^t)^2 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} \sum_t r^t \\ \sum_t r^t x^t \end{bmatrix}$$

$$\mathbf{w} = \mathbf{A}^{-1} \mathbf{y}$$

Polynomial Regression

$$g(x^t | w_k, \dots, w_2, w_1, w_0) = w_k (x^t)^k + \dots + w_2 (x^t)^2 + w_1 x^t + w_0$$

$$\mathbf{D} = \begin{bmatrix} 1 & x^1 & (x^1)^2 & \dots & (x^1)^k \\ 1 & x^2 & (x^2)^2 & \dots & (x^2)^k \\ \vdots & & & & \\ 1 & x^N & (x^N)^2 & \dots & (x^N)^k \end{bmatrix} \quad \mathbf{r} = \begin{bmatrix} r^1 \\ r^2 \\ \vdots \\ r^N \end{bmatrix}$$

$$\mathbf{w} = (\mathbf{D}^T \mathbf{D})^{-1} \mathbf{D}^T \mathbf{r}$$

Other Error Measures

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□ Square Error: $E(\theta | \mathcal{X}) = \frac{1}{2} \sum_{t=1}^N [r^t - g(x^t | \theta)]^2$

□ Relative Square Error: $E(\theta | \mathcal{X}) = \frac{\sum_{t=1}^N [r^t - g(x^t | \theta)]^2}{\sum_{t=1}^N [r^t - \bar{r}]^2}$

□ Absolute Error: $E(\vartheta | X) = \sum_t |r^t - g(x^t | \vartheta)|$

□ ε -sensitive Error:

$$E(\vartheta | X) = \sum_t 1(|r^t - g(x^t | \vartheta)| > \varepsilon) (|r^t - g(x^t | \vartheta)| - \varepsilon)$$

Bias and Variance

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$$E[(r - g(x))^2 | x] = \underbrace{E[(r - E[r | x])^2 | x]}_{\text{noise}} + \underbrace{(E[r | x] - g(x))^2}_{\text{squared error}}$$

$$E_x[(E[r | x] - g(x))^2] = \underbrace{(E[r | x] - E_x[g(x)])^2}_{\text{bias}} + \underbrace{E_x[(g(x) - E_x[g(x)])^2]}_{\text{variance}}$$

Estimating Bias and Variance

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- M samples $X_i = \{x_i^t, r_i^t\}, i = 1, \dots, M$
are used to fit $g_i(x), i = 1, \dots, M$

$$\text{Bias}^2(g) = \frac{1}{N} \sum_t [\bar{g}(x^t) - f(x^t)]^2$$

$$\text{Variance}(g) = \frac{1}{NM} \sum_t \sum_i [g_i(x^t) - \bar{g}(x^t)]^2$$

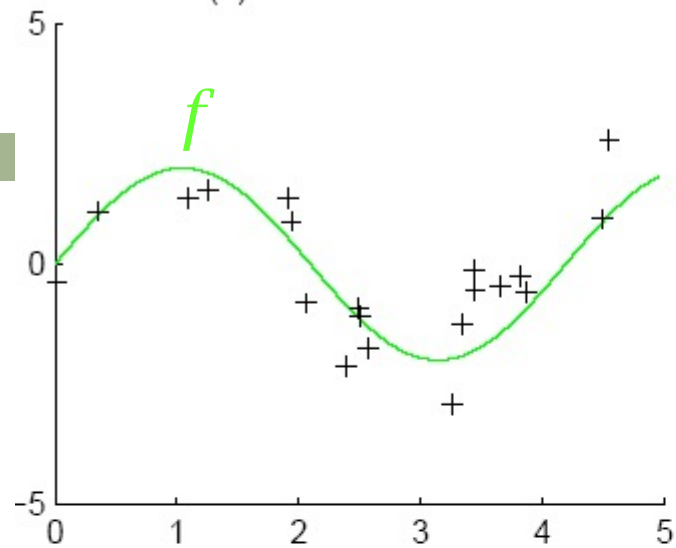
$$\bar{g}(x) = \frac{1}{M} \sum_t g_i(x)$$

Bias/Variance Dilemma

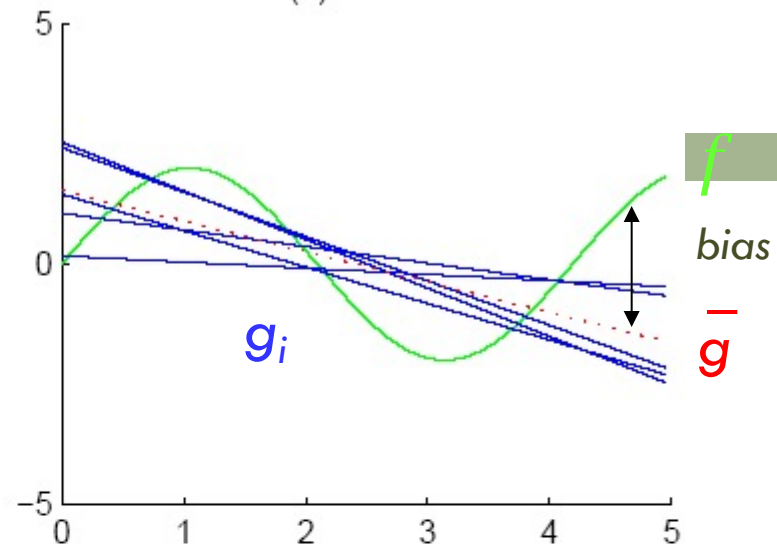
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- Example: $g_i(x)=2$ has no variance and high bias
 $g_i(x)=\sum_t r_i^t/N$ has lower bias with variance
- As we increase complexity,
 bias decreases (a better fit to data) and
 variance increases (fit varies more with data)
- Bias/Variance dilemma: (Geman et al., 1992)

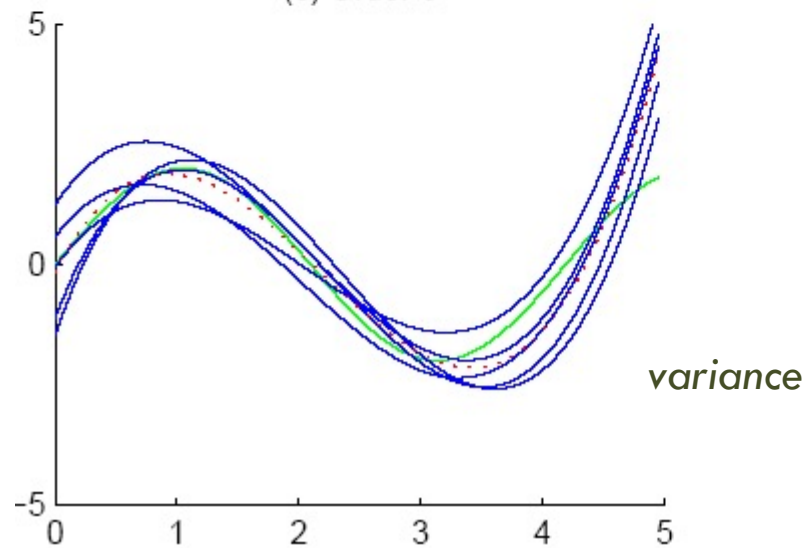
(a) Function and data



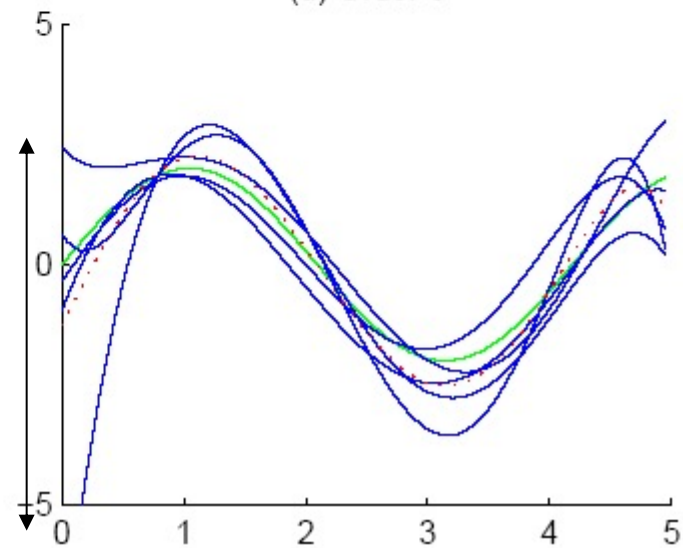
(b) Order 1



(c) Order 3

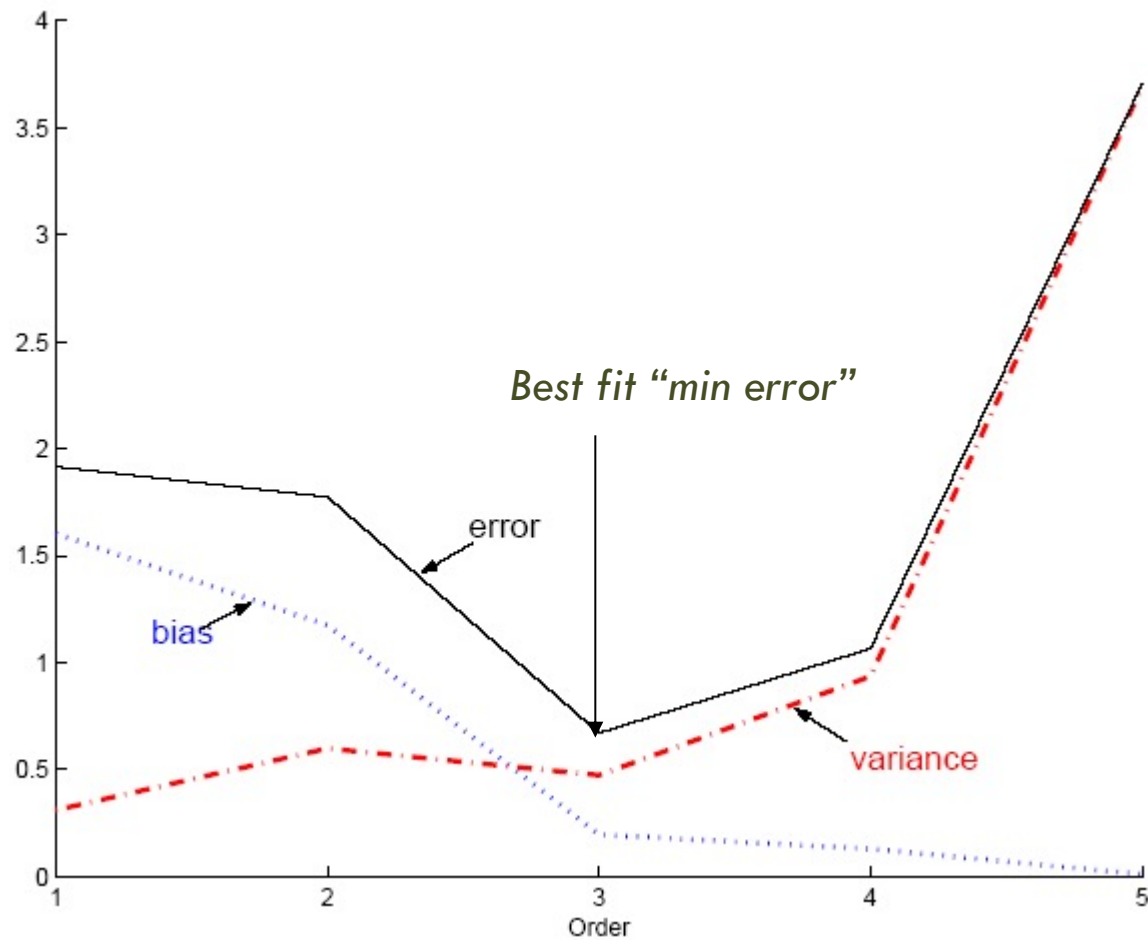


(d) Order 5

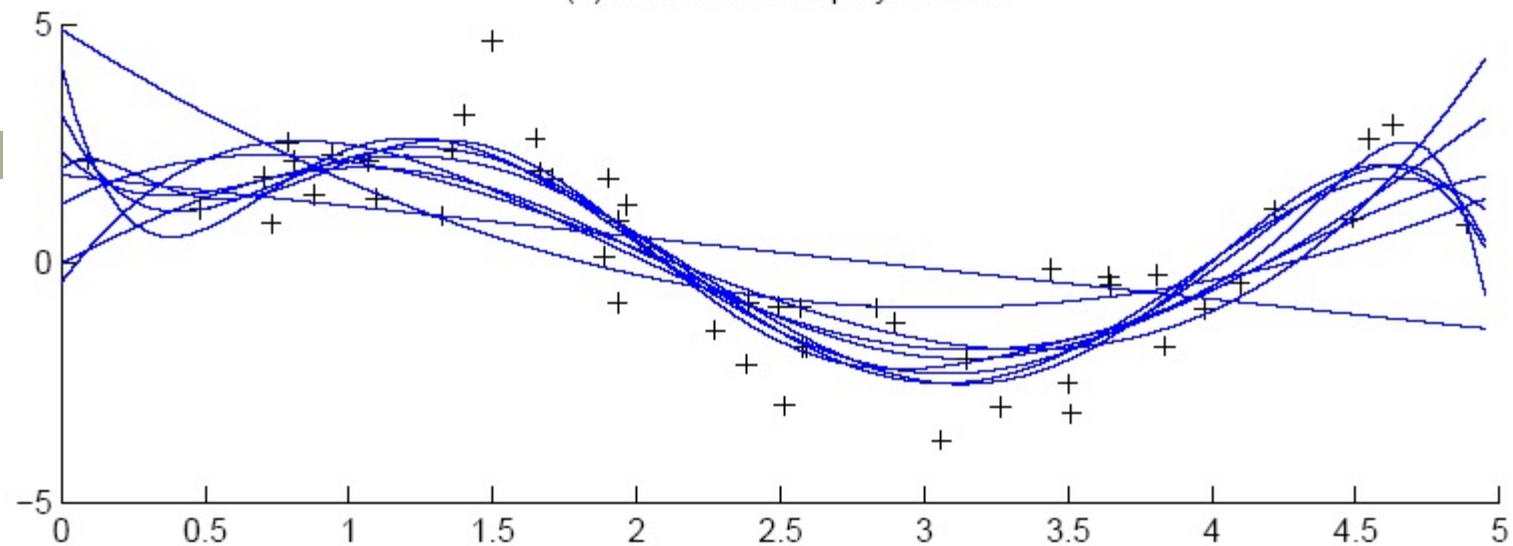


Polynomial Regression

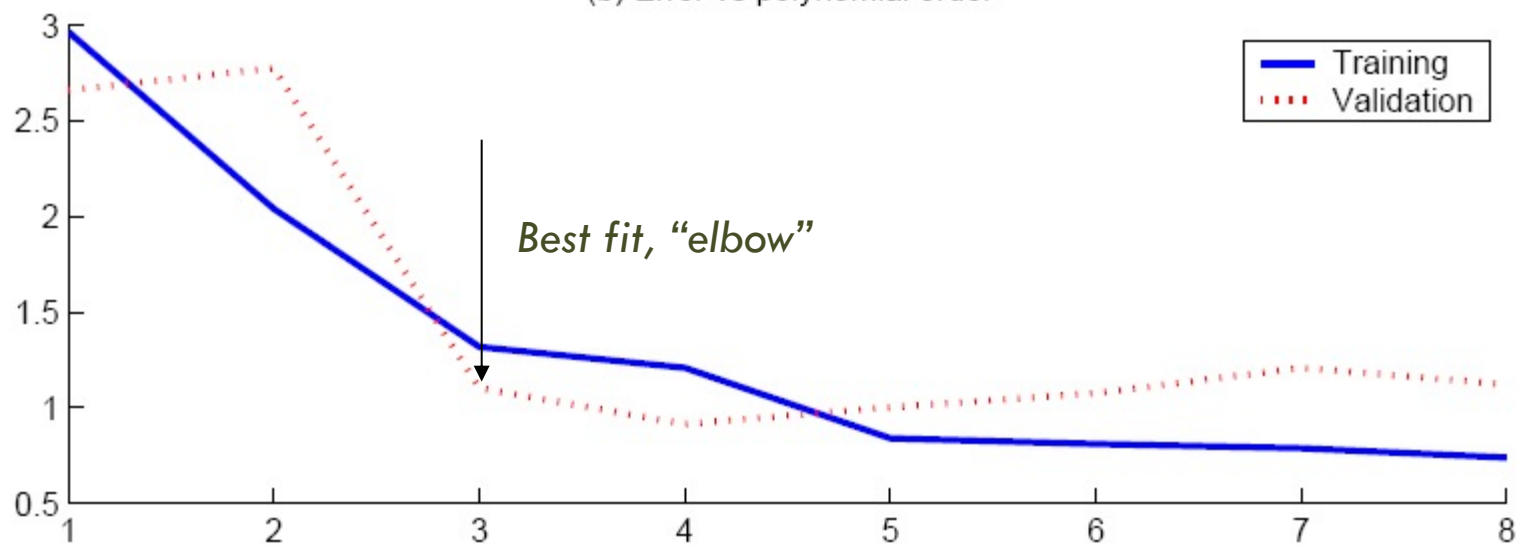
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(a) Data and fitted polynomials



(b) Error vs polynomial order



Model Selection

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- Cross-validation: Measure generalization accuracy by testing on data unused during training
- Regularization: Penalize complex models
 $E' = \text{error on data} + \lambda \text{ model complexity}$
Akaike's information criterion (AIC), Bayesian information criterion (BIC)
- Minimum description length (MDL): Kolmogorov complexity, shortest description of data
- Structural risk minimization (SRM)

Bayesian Model Selection

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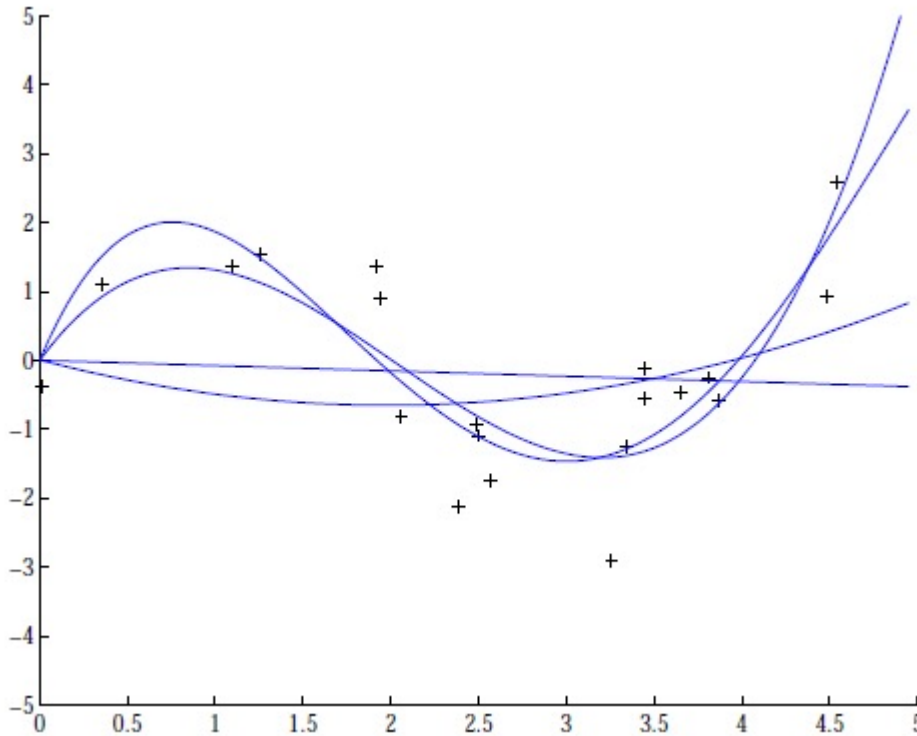
- Prior on models, $p(\text{model})$

$$p(\text{model} | \text{data}) = \frac{p(\text{data} | \text{model}) p(\text{model})}{p(\text{data})}$$

- Regularization, when prior favors simpler models
- Bayes, MAP of the posterior, $p(\text{model} | \text{data})$
- Average over a number of models with high posterior (voting, ensembles: Chapter 17)

Regression example

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Coefficients increase in magnitude as order increases:

1: [-0.0769, 0.0016]

2: [0.1682, -0.6657, 0.0080]

3: [0.4238, -2.5778, 3.4675, -0.0002]

4: [-0.1093, 1.4356, -5.5007, 6.0454, -0.0019]

Regularization (L2):
$$E(\mathbf{w} | \mathcal{X}) = \frac{1}{2} \sum_{t=1}^N [r^t - g(x^t | \mathbf{w})]^2 + \lambda \sum_i w_i^2$$